

YOLO Model to detect Object.

Aim:-

To implement YOLO object detection model for real time detection and localization of multiple objects in an image or video

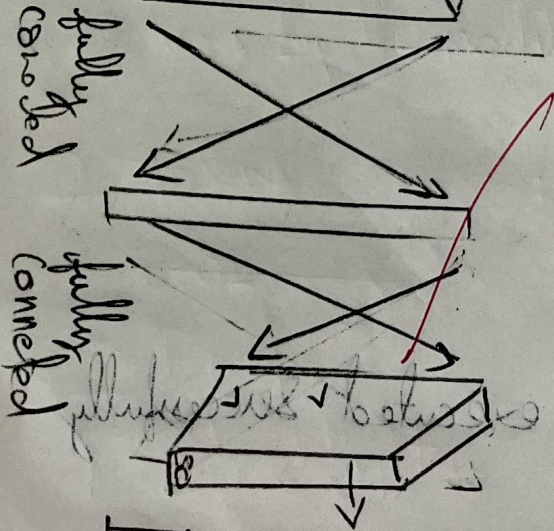
Objective:-

- ① To understand the working principle and architecture of the YOLO object detection model
- ② To use a pre-trained YOLO model for detecting multiple objects in real world images
- ③ To visualize bounding boxes, class labels, and confidence scores for detected objects
- ④ To experiment with the test images and ~~show~~ YOLO's real-time performance
- ⑤ To analyse YOLO's advantages in speed and accuracy compared to traditional detection methods

Structure

- ① Add new context layers for classification
- ② Rewrite the context model with changing registers like below
- ③ To provide better performance
- ④ Freeze all convolutional layers
- ⑤ Load pre-trained CNN model
- ⑥ Load dataset and preprocess it
- ⑦ Import necessary libraries

END



Deletions: 98 per class



non-Maximum Suppression

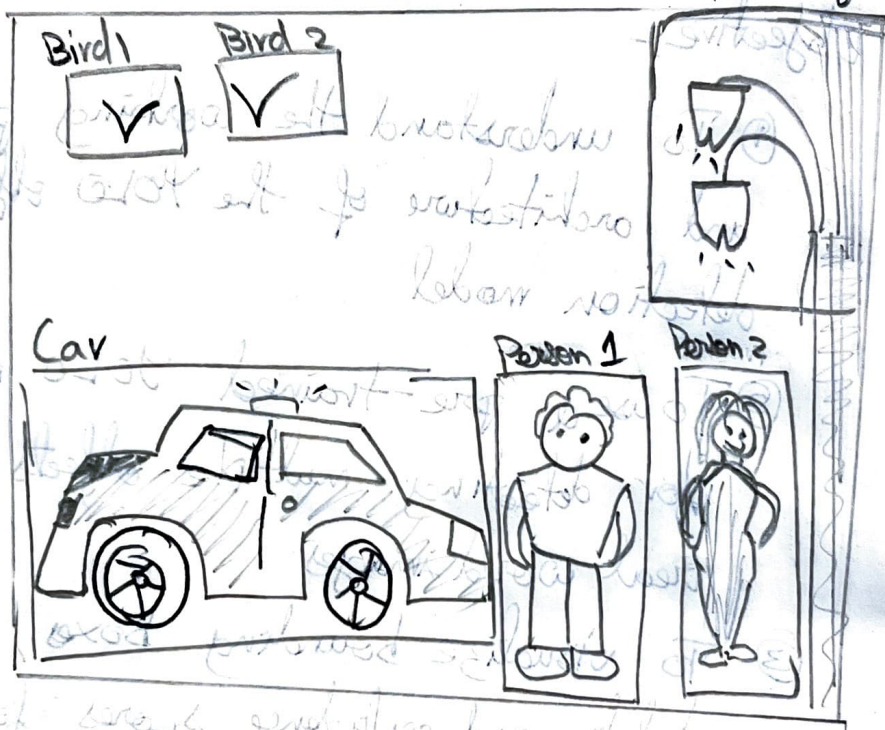
testbed testbed of labat old

Person 1

labat testbed testbed old transmission of
networks have networked with low ref
radio volume: no ni stable shithun je

Observation:

Street Light



Detections by the model

2 persons

: 0.93 ; 0.85

1 Car

: 0.98

2 Bird

: 0.62 ; 0.64

~~Street Light~~

: 0.82

Pseudocode :

BEGIN

- ① Import necessary libraries
- ② Load a pre-trained model
eg: YOLO v8 from Ultralytics
- ③ Input an image or capture video frame
- ④ Perform object detection using the model
- ⑤ For each detected object
 Extract bounding box coordinates,
 class label, and confidence score

 Draw bounding box and label on
 the image
- ⑥ Display the output image with
 detections

~~END~~

Result:-

The code executed successfully